HW 10 Solutions 10. 19, 20, 22, 35, 40, 62, 68, 69, 74

10.19 (a) $V_{cm} = R\omega = (0.600\,m)(3.00\,rad/s) = 1.80\,m/s$

(b) $K = \frac{1}{2}MV_{cm}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}MV_{cm}^2 + \frac{1}{2}(MR^2)\left(\frac{V_{cm}^2}{R^2}\right) = MV_{cm}^2$

$K = (2.20\,kg)(1.80\,m/s)^2 = 7.13\,J$

(c) (i) $V = 2V_{cm} = 3.60\,m/s$, $\vec{v}$ is to the right (ii) $V=0$

(iii) $V = \sqrt{V_{cm}^2 + V_{tan}^2} = \sqrt{V_{cm}^2 + (R\omega)^2} = \sqrt{2}\,V_{cm} = 2.55\,m/s$. $\vec{v}$ at this point is at 45° below the horizontal.

(d) To someone moving to the right at $V = V_{cm}$, the hoop appears to rotate about a stationary axis at its center.

(i) $V = R\omega = 1.80\,m/s$, to the right

(ii) $V = 1.80\,m/s$, to the left

(iii) $V = 1.80\,m/s$, downward

10.20 (a) Conservation of energy gives $U_i = K_f$. $U_i = Mgy_i$

$K_f = \frac{1}{2}MR^2\omega^2 + \frac{1}{2}(MR^2)\omega^2 = MR^2\omega^2$, so $Mgy_i = MR^2\omega^2$

$\omega = \frac{\sqrt{gy_i}}{R} = \frac{\sqrt{(9.80\,m/s^2)(0.750\,m)}}{0.0800\,m} = 33.9\,rad/s$

(b) $V = R\omega = (0.0800\,m)(33.9\,rad/s) = 2.71\,m/s$

10.22 (a) $\Sigma F_x = ma_x$ gives $mg\sin\beta - f = ma_{cm}$. $\Sigma\tau_z = I\alpha_z$ gives $fR = (\frac{2}{3}mR^2)\alpha$.

$\alpha = \frac{a_{cm}}{R}$, so $f = \frac{2}{3}ma_{cm}$

Combining the equations gives $mg\sin\beta = \frac{5}{3}ma_{cm}$

$a_{cm} = \frac{3}{5}g\sin\beta = \frac{3}{5}(9.80\,m/s^2)(\sin 38.0°) = 3.62\,m/s^2$

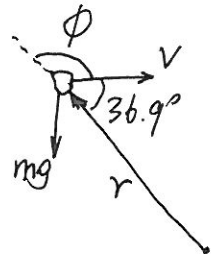
$f = \frac{2}{3}ma_{cm} = \frac{2}{3}(2.00\,kg)(3.62\,m/s^2) = 4.83\,N$.

The friction is static since there is no slipping at the point of contact.

$n = mg\cos\beta = 15.45\,N$. $\mu_s = \frac{f}{n} = \frac{4.83\,N}{15.45\,N} = 0.313$

(b) The acceleration is independent of m and doesn't change. The friction force is proportional to m so will double; $f = 9.66\,N$. The normal force will also double, so the minimum μ_s required for no slipping wouldn't change.

10.35



(a) $L = mvr\sin\phi$

$= (2.00\,kg)(12.0\,m/s)(8.00\,m)\sin 143.1°$

$L = 115\ kg\cdot m^2/s$

To find the direction of $\vec{L}$ apply the right-hand rule by turning $\vec{r}$ into the direction of $\vec{v}$ by pushing on it with the fingers of your right hand. $\vec{L}$ points into the page.

(b) $\tau = mgr\cos 36.9° = 125\ kg\cdot m^2/s^2$

To find the direction of $\vec{\tau}$ and hence of $\frac{d\vec{L}}{dt}$ apply the right-hand rule. $\frac{d\vec{L}}{dt}$ points out of the page.

10.40 (a) Yes, angular momentum is conserved. The moment arm for the tension in the cord is zero so this force exerts no torque and there is no net torque on the block.

(b) $L_1 = L_2$ so $I_1\omega_1 = I_2\omega_2$. Block treated as a point mass, so $I = mr^2$

$mr_1^2\omega_1 = mr_2^2\omega_2$

$\omega_2 = \left(\frac{r_1}{r_2}\right)^2\omega_1 = \left(\frac{0.300\,m}{0.150\,m}\right)^2(1.75\,rad/s) = 7.00\,rad/s$

(c) $K_1 = \frac{1}{2}mv_1^2 = \frac{1}{2}I_1\omega_1^2 = \frac{1}{2}mr_1^2\omega_1^2$

$K_1 = \frac{1}{2}(0.0250\,kg)(0.300\,m)^2(1.75\,rad/s)^2 = 0.00345\,J$

$K_2 = \frac{1}{2}mv_2^2 = \frac{1}{2}I_2\omega_2^2 = \frac{1}{2}mr_2^2\omega_2^2$

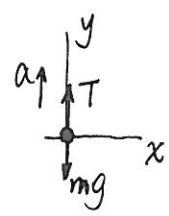
$K_2 = \frac{1}{2}(0.0250\,kg)(0.150)^2(7.00\,rad/s)^2 = 0.01378\,J$

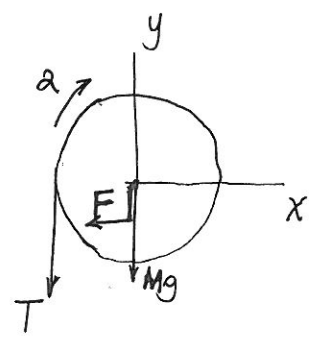
$\Delta K = K_2 - K_1 = 0.0103\,J$

(d) $W_{tot} = \Delta K$

But $W_{tot} = W$, the work done by the tension in the cord, so $W = 0.0103\,J$

10.62 For the crate





$\Sigma F_y = ma_y \qquad T - mg = ma$

$T = m(g+a) = 50\,kg\,(9.80\,m/s^2 + 0.80\,m/s^2)$

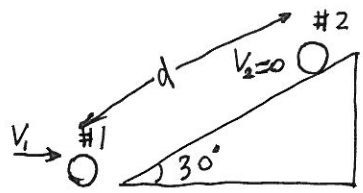
$= 530\,N$

$\Sigma\tau_z = I\alpha_z \qquad Fl - TR = I\alpha_z$

$a = R\alpha_z$ so $\alpha_z = \frac{a}{R} \qquad Fl - TR = I\frac{a}{R}$

$F = T\left(\frac{R}{l}\right) + \frac{Ia}{Rl} = 530\,N\left(\frac{0.25\,m}{0.12\,m}\right) + \frac{(2.9\,kg\cdot m^2)(0.80\,m/s^2)}{(0.25\,m)(0.12\,m)} = 1200\,N$

10.68 (a) Conservation of energy



$K_1 + U_1 + W_{other} = K_2 + U_2$

$U_1 = Mgy_1 = 0$, $W_{other} = 0$

$K_1 = \frac{1}{2}MV_1^2 + \frac{1}{2}I_{cm}\omega_1^2$, But $\omega_1 = \frac{V_1}{R}$ and $I_{cm} = \frac{1}{2}MR^2$

so $K_1 = \frac{1}{2}MV_1^2 + \frac{1}{2}(\frac{1}{2}MR^2)(\frac{V_1}{R})^2 = \frac{3}{4}MV_1^2$

$U_2 = Mgy_2 = Mgd\sin 30°$

$K_2 = 0$ (disk is at rest at point 2)

Thus $\frac{3}{4}MV_1^2 = Mgd\sin 30°$

$$d = \frac{3V_1^2}{4g\sin 30°} = \frac{3(2.50\,m/s)^2}{4(9.80\,m/s^2)\sin 30°} = 0.957\,m$$

(b) The mass M enters both in the linear inertia and in the gravity force so divides out. The mass M and radius R enter in both the rotational inertia and the gravitational torque so divide out.

10.69 $I = 2(\frac{1}{2}mR^2) = mR^2$. The moment arm for T is b.
The tension is related to the acceleration of the yo-yo by $(2m)g - T = (2m)a$, and to the angular acceleration by $Tb = I\alpha = I\frac{a}{b}$.
Combining these equations we get

$$a = g\frac{2m}{2m + I/b^2} = g\frac{2}{2+(R/b)^2}, \quad \alpha = g\frac{2}{2b + R^2/b}$$

The tension

$$T = 2m(g-a) = (2mg)\left(1 - \frac{2}{2b - R^2/b}\right) = \frac{2mg}{2(b/R)^2+1}$$

10.74 (a) Projectile motion: $V_{oy} = 0$. $a_y = 9.80\,m/s^2$. $y - y_0 = 20\,m$.
$y - y_0 = V_{oy}t + \frac{1}{2}a_y t^2$ gives $t = \sqrt{\frac{2(y-y_0)}{a_y}} = 2.02\,s$.
Then $x - x_0 = V_{ox}t$ gives $V = V_{ox} = \frac{x - x_0}{t} = \frac{36\,m}{2.02\,s} = 17.8\,m/s$
Motion down the hill: $U_1 = K_f$. $mgh = \frac{7}{10}mV^2$. $h = \frac{7V^2}{10g} = \frac{7(17.8\,m/s)^2}{10(9.80\,m/s^2)} = 22.6\,m$

(b) $\frac{1}{2}I\omega^2 = \frac{1}{5}mv^2$, independent of R. I is proportional to R^2 but ω^2 is proportional to $\frac{1}{R^2}$ for a given translational speed v.

(c) The object still needs $V = 17.8\,m/s$ at the bottom of the hill in order to clear the pit. But now $K_f = \frac{1}{2}mv^2$ and $h = \frac{V^2}{2g} = 16.2\,m$.